

Process control methods for fin-FET gate uniformity at sub-7-nanometre nodes

US9876543 · Macrosilicon Manufacturing Corp.

PSS 78.91

S

4 / 183 · Top 2.2%

STATUS

Active

OWNER

Macrosilicon Manufacturing Corp.

IPC NUMBER

H01L 21/336

FILED

2018-06-04

CLAIMS

32 (4 indep · 28 dep)

FILING JURISDICTION

US

FAMILY COUNTRIES

US · TW · KR · EP · CN · JP

IPC CATEGORY

Semiconductor process — gate fabrication

GRANTED

2021-09-14

REMAINING LIFE

~12 yrs · until 2038-06-04

COHORT PERCENTILE

97.8%

183-patent semiconductor advanced-node process pool

RANK IN POOL

4 / 183

Re-ranked weekly as cohort updates

PSS SCORE

78.91

Composite of 8 pillars · 50 indicators

INDICATORS GRADED

50 / 50

COHORT SIZE

183 patents

SUMMARY

A wafer-level process-control loop that ties **real-time electron-beam metrology** to gate-etch endpoint detection, materially tightening gate-CD uniformity at the 5–7 nm node. Applied during fin-FET fabrication, it cuts cross-wafer CD variance and improves yield in high-mobility-channel architectures.

COHORT NUMBER

H01L

COHORT NAME

Semiconductor advanced-node process

FAMILY JURISDICTIONS

US · TW · KR · EP · CN · JP

RATING RATIONALE

Top 2.2% in its industry pool. **Market Coverage (P4 · 86.50)** and **Citation Impact (P1 · 84.20)** are the rating anchors — full six-jurisdiction footprint and a top-decile citation tail in advanced-node process IP. Stable across all five scenarios at 96%+; ranked #4 on the anchor scenario. Forward-Looking Value (P8 · 68.50) is the softest pillar but still well above pool median. Read as **a portfolio anchor of the highest grade** — the kind of asset an IC memo opens with.

How is this calculated?

Pillars & Indicators

Eight pillars · 50 indicators graded — how the patent strength score is decomposed.

PILLAR ANALYSIS

The Patent Strength Score is a weighted composite of eight pillars, each graded against 183 peers in the cohort.

PILLAR	SCORE	PROFILE VS COHORT MEDIAN	READ
P1 Citation Impact	84.20		KEY DRIVER
P2 Legal Strength	81.40		KEY DRIVER
P3 Technical Value	77.30		SUPPORTIVE
P4 Market Coverage	86.50		KEY DRIVER
P5 Text Quality	72.10		SUPPORTIVE
P6 Network Centrality	81.80		KEY DRIVER
P7 Licensing Potential	79.40		SUPPORTIVE
P8 Forward-Looking Value	68.50		SUPPORTIVE

≥ 70 · TOP QUARTILE

35–70 · MID BAND

< 35 · BELOW MEDIAN

| COHORT MEDIAN (50)

PILLAR DEEP DIVE

Each pillar decomposes into 4–8 individual indicators. Percentiles below are in-pool — derived against the 183-patent semiconductor advanced-node process cohort.

Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6	Pillar 7	Pillar 8
Citation Impact	Legal Strength	Technical Value	Market Coverage	Text Quality	Network Centrality	Licensing Potential	Forward-Looking Value
84.20	81.40	77.30	86.50	72.10	81.80	79.40	68.50

CODE	INDICATOR	VALUE	PERCENTILE	WEIGHT
P1.1	Forward citations	47	99.5 %	20%
P1.2	Citation velocity	4.17/yr	98.9 %	20%
P1.3	Generality index	0.87	91.5 %	15%
P1.4	Originality index	0.77	84.2 %	10%
P1.5	Disruption index (CD)	0.24	76.9 %	15%
P1.6	Backward citations	19	69.5 %	10%
P1.7	Examiner-added citation ratio	0.66	62.2 %	5%

Citation Impact is reading **top-quartile**. Forward citations and velocity both top-decile. The drivers concentrated here anchor the rating.

IMPORTANT DISCLOSURES

For diligence use only. Brief reports synthesize indicators from the Innovue 170M-patent index. They are not legal opinions, not infringement analyses, and do not replace counsel review. Tier and percentile are pool-relative, not absolute. Pool composition is documented in [Methodology v0.5](#).

Cohort & refresh. The 183-patent semiconductor advanced-node process pool is re-ranked weekly as new patents enter and old ones lapse. Rank, percentile, and PSS may move week-over-week even when the patent's own indicators do not.

Scoring transparency. Of 50 indicators, 34 are computed from structured data and 16 are graded by NLP. Pillar weights are documented in [Methodology v0.5](#); raw indicator values are shown in Pillar Deep Dive. Risk & Action — drivers, scenarios, recommended actions — is available in the Pro edition.

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PRO ADDS

Risk & Action — what Pro unlocks

Key Rating Drivers

Top-10 strengths and weaknesses with sensitivity sliders.

Scenario Sensitivity

Five-scenario stress band with reweighted PSS bounds.

Action Considerations

Do / mind / watch grid tailored to your tier.

Full Disclosures

Source attribution, pool composition, methodology notes.